EE1101

Quiz 1

Disclaimer:

This paper has been built by reverse-engineering the answer paper after evaluation. The questions are only approximate replicas of the original paper.

1. Find and plot the response when a system with impulse response $e^{-\alpha|t|}$ is excited by $u(t)$.
   (5 marks)

2. Let a periodic signal with time period $T = 4$ consist for a triangular pulse of amplitude B with a DC offset of 1 for half of the time period and remain constant at 1 for the rest of the period.
   If the power of this signal is found to be $\frac{5}{3}$, find B.
   (5 marks)

3. Consider a discrete time system which when given an input $x[n]$, outputs $x[n]$ when n is even and outputs 0 when n is odd.
   Verify whether the system is Linear, Time Invariant, Stable, Causal and Invertible.
   (5 marks)

4. Let $x(t) = \begin{cases} t & 0 \leq x \leq 1 \\ -1 & 1 \leq x \leq \frac{3}{2} \\ 0 & otherwise \end{cases}$
   Plot
   a) odd part of $x(t)$ **(2 marks)**
   b) $x(5-2t)$ **(3 marks)**

1. Find and plot the response when a system with impulse response $e^{-\alpha|t|}$ is excited by $u(t)$.

(5 marks)

$h(t) = e^{-\alpha|t|} \qquad x(t) = u(t)$

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} e^{-\alpha|\tau|}\, u(t-\tau)\, d\tau$$

$$= \int_{t}^{\infty} 0\, d\tau + \int_{-\infty}^{t} e^{-\alpha|\tau|}\, 1\, d\tau$$

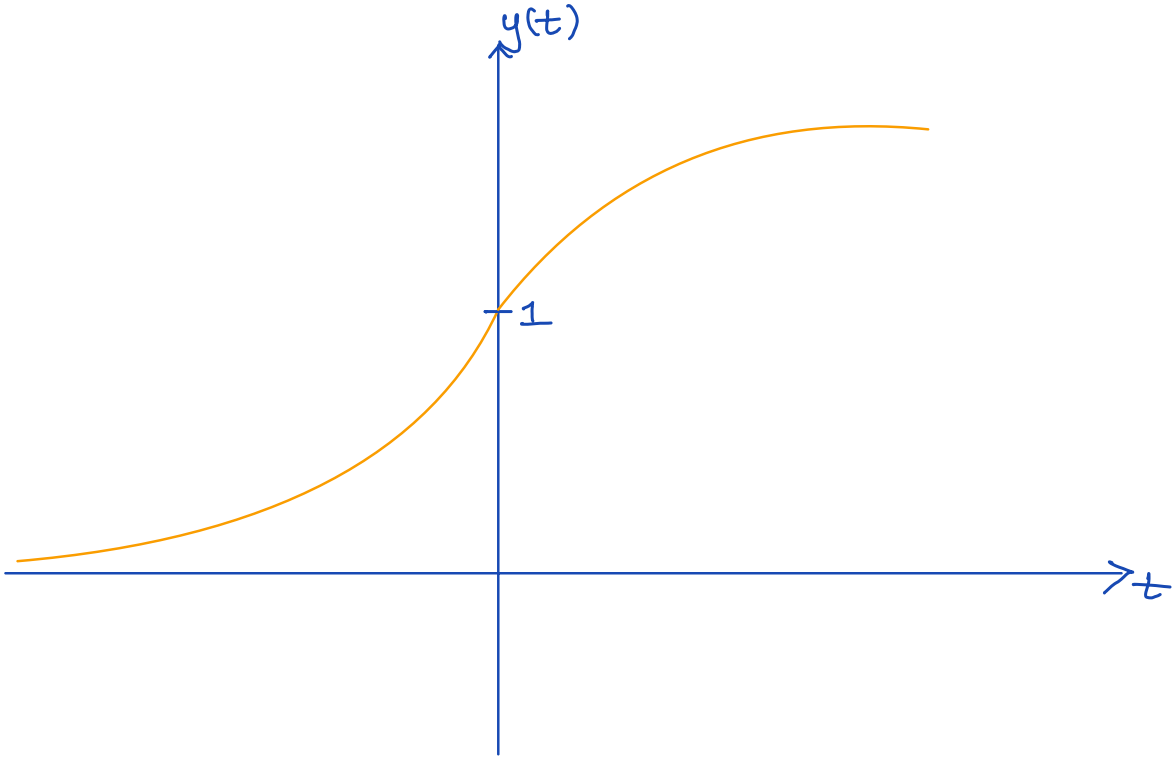
$$\int_{-\infty}^{0} e^{\alpha\tau}\, d\tau + \int_{0}^{t} e^{-\alpha\tau}\, d\tau$$

$$= \frac{1}{\alpha}[1] + \frac{-1}{\alpha}\left[e^{-\alpha t} - 1\right]$$

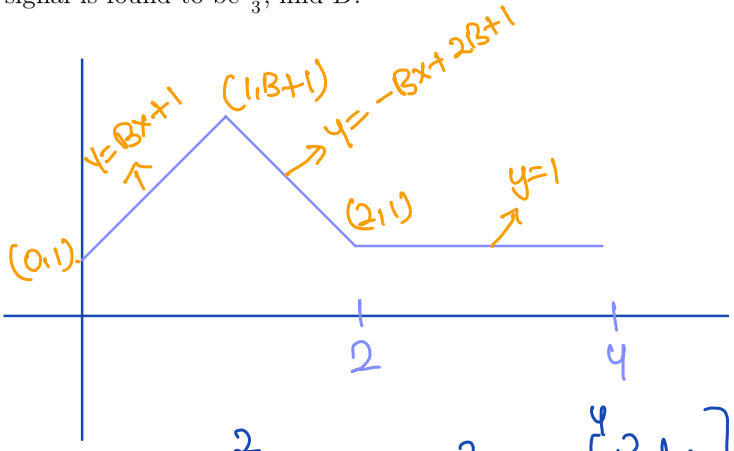
$$= \frac{2 - e^{-\alpha t}}{\alpha}$$

$t > 0$: $y(t) = \dfrac{2 - e^{-\alpha t}}{\alpha}$

$t < 0$: $y(t) = \dfrac{e^{\alpha t}}{\alpha}$



2. Let a periodic signal with time period $T = 4$ consist for a triangular pulse of amplitude B with a DC offset of 1 for half of the time period and remain constant at 1 for the rest of the period.
If the power of this signal is found to be $\frac{5}{3}$, find B.
(5 marks)



$$\frac{1}{4}\left[\int_0^1 (Bx+1)^2 dx + \int_1^2 (-Bx+2B+1)^2 dx + \int_2^4 1^2 dx\right] = 5/3$$

$$\frac{B^2}{3} + B + 1 + \frac{7B^2}{3} + 4B^2 + 4B + 1 + (2B+1)(-B)(3) + 2 = \frac{20}{3}$$

$$\frac{2B^2}{3} + 2B = 8/3 \Rightarrow B^2 + 3B = 4$$

$$\Rightarrow B = 1, -4$$

3. Consider a discrete time system which when given an input $x[n]$, outputs $x[n]$ when n is even and outputs 0 when n is odd.
Verify whether the system is Linear, Time Invariant, Stable, Causal and Invertible.
(5 marks)

Linear ✓

Time invariant ✗

Stable ✓

Causal ✓

Invertible ✗

4. Let $x(t) = \begin{cases} t & 0 \le x \le 1 \\ -1 & 1 \le x \le \frac{3}{2} \\ 0 & otherwise \end{cases}$

Plot

a) odd part of $x(t)$ **(2 marks)**

b) $x(5-2t)$ **(3 marks)**

